

Bluetooth Channel Sounding

Inline Phase Transfer

Returning the reflector's phase in the radio signal instead of the data path: the mechanism, the transmitter it demands, what it costs, what it removes, and when to enable it

Inline PCT Transfer — IPT, and the subject of this report — is an optional Channel Sounding capability in which the reflector pre-rotates its outgoing tone by the phase it has just measured, so that the initiator's own single-sided measurement already carries the full two-way channel phase. The reflector's phase then no longer has to be transported anywhere: it has been returned *inline*, in the radio signal itself, and the phase correction term the reflector reports is left with zero imaginary part, conveying amplitude alone. This report derives that mechanism from the specification's own phase algebra, shows why the construction is exactly phase-neutral, and then examines what it actually demands of a reflector: a transmitter whose carrier phase can be commanded per antenna path, to a resolution that this report quantifies, and settled inside a switch period, with the whole measure–compute–apply loop closed in as little as fifteen microseconds. It shows that IPT is thermally free — the two-way phase variance is unchanged — and that its real cost is a new hardware term and the permanent loss of the reflector's phase as an auditable quantity. It closes with the negotiation and configuration path, an error budget, a decision rule, and a verification plan.

Primary references Core 6.3, Vol. 6, Part A, Sec. 6.5; Vol. 1, Part A, Sec. 9.2

Step behaviour Core 6.3, Vol. 6, Part H, Secs. 4.3.3, 4.3.4 and 4.6

Capability and
configuration Core 6.3, Vol. 4, Part E; Vol. 6, Part B

Companion reports *Mode-0 Frequency and Timing Calibration*, Rev. 1.0;
Mode-1 RTT Estimation, Rev. 1.0; *Mode-2 Phase-Based
Ranging*, Rev. 1.0; *Mode-3 Combined RTT and PBR*,
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Abstract

In ordinary phase-based ranging the two devices each measure the phase of the other's tone, each reports a phase correction term, and a ranging entity multiplies the two complex terms together. The product cancels the unknown relative local-oscillator phase and leaves twice the channel phase. That construction requires both measurements to reach the same place, which means the reflector's phase must travel back over some data path before any distance exists.

Inline PCT Transfer removes that requirement. With IPT enabled, the reflector applies the phase it has just measured to its own outgoing carrier [Core 6.3, Vol. 6, Part A, Sec. 6.5]. The relative phase the initiator then measures at its antenna is $\theta_{INIT}(f) = \theta_{CH}(f) - \Delta\theta_{LO}(f) + \theta_{REFL}(f) = 2\theta_{CH}(f)$ [Core 6.3, Vol. 1, Part A, Sec. 9.2]: the local-oscillator term cancels in the air rather than in arithmetic, and the initiator holds the complete two-way observable from its own measurement alone. The reflector's reported term becomes $PCT_{REFL}(f) = A_{REFL}(f)e^{i0}$ — the specification requires the imaginary part to be zero and the real part to carry the measured amplitude — so the channel transfer function assembled by the ranging entity is unchanged.

This report treats IPT as an engineering trade rather than a feature. Three findings frame it. It is *thermally free*: the two-way phase error is the sum of the same two independent phase estimates as before, so the variance is identical and no precision is bought or lost. Its real cost is a *new hardware term*: the reflector must command its carrier phase per antenna path, and this report shows that eight bits of applied-phase resolution costs 2.5% of ranging precision while six bits costs 34%. And it is *irreversible*: because the correction is applied in the radio rather than recorded, the reflector's phase estimate can never be re-examined, re-processed or audited, and the diagnostic separation between initiator-side and reflector-side error is gone for good. The report quantifies the real-time budget the reflector must close — as little as $15\mu s$ between measuring a path and transmitting it — explains the separate T_{SW_IPT} and T_{IP2} capability declarations that exist because of it, and argues that IPT pairs naturally with Mode-3, whose round-trip time restores the independent check that IPT removes.

Status and interpretation. Normative claims are tied to the cited Bluetooth Core Specification sections and have been checked against the adopted Core 6.3 text. The error analysis, the applied-phase resolution study, the real-time budget, the decision rule and the verification plan are this report's own and are not specified. The Bluetooth specification and adopted test specifications remain authoritative for conformance.

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1 Scope, naming, and what IPT changes

1.1 Naming

The specification uses two names for one capability. Volume 1, Part A calls it *Channel Sounding Inline Phase Correction Term Transfer*; Volume 6, Part A, Section 6.5 calls it *Inline PCT Transfer* and abbreviates it IPT, which is the form used in the link-layer and host-interface parameter names. It is colloquially called *inline phase-based ranging* or *inline PBR*, because its effect is to complete a phase-based ranging measurement inside the radio exchange. This report uses IPT throughout.

IPT is a *reflector* capability. It is declared by the reflector, enabled by configuration, and it changes what the reflector transmits and what it reports; the initiator's own measurement and reporting format are untouched [Core 6.3, Vol. 4, Part E].

1.2 The problem it solves

The companion Mode-2 report develops the ordinary construction. Each device measures the phase of the other's tone against its own oscillator; each reports a phase correction term; the ranging entity multiplies the two complex terms, and the unknown relative oscillator phase cancels in the product. The construction is exact and needs no shared time base — but it needs both halves in one place.

That requirement is the whole motivation for IPT. Until the reflector's phase correction terms reach whatever computes the range, no distance exists. In a system where the initiator is the device that needs the answer, those terms must traverse the reflector's controller, its host, and the connection back to the initiator, for every tone on every channel. The measurement is finished in microseconds; the answer waits on a data path.

What IPT moves. It does not make the measurement better and it does not make it faster in the radio. It moves *where the two-way phase is formed* — out of the ranging entity's arithmetic and into the reflector's transmitter. The consequence is that the initiator's own measurement is already the answer, so the ranging-critical half of the data no longer has to be carried anywhere.

1.3 What this report covers

Sections 3 and 4 derive the mechanism and the transmitter it demands. Section 5 covers the timing parameters that exist only because of IPT. Section 6 covers the altered reporting. Section 7 covers capability and configuration. Sections 8–9 are the engineering assessment: what changes in the error budget, and what is permanently given up. Sections 10–11 cover integrity and the decision to enable it. The remainder follows the pattern of the companion reports.

2 Normative framework and terminology

Table 1: Normative map for Inline PCT Transfer.

Source	What it defines	Question answered
Core 6.3, Vol. 1, Part A, Sec. 9.2	The phase algebra, with and without IPT	Why does the initiator's own measurement suffice?
Core 6.3, Vol. 6, Part A, Sec. 6.5	IPT normative behaviour: $Q = 0$, phase accuracy, carrier adjustment	What must the reflector do?
Core 6.3, Vol. 6, Part A, Sec. 6.2	Phase measurement accuracy requirements, which still apply	To what accuracy?
Core 6.3, Vol. 6, Part H, Sec. 4.3.3	Mode-2 step; T_{SW_IPT} ; tone extension slot exemption under IPT	How does the step change?
Core 6.3, Vol. 6, Part H, Sec. 4.3.4	Mode-3 step, to which IPT equally applies	Does IPT work with Mode-3?
Core 6.3, Vol. 6, Part H, Sec. 4.6	PCT encoding, reference power level, tone quality	How is the result reported?
Core 6.3, Vol. 6, Part B	LL_CS_CAPABILITIES exchange and configuration defaults	How is IPT negotiated?
Core 6.3, Vol. 4, Part E	Subfeatures_Supported bit 4, CS_Enhancements bit 0, T_{IP2}/T_{SW_IPT} time lists	How is it declared and enabled?

Table 2: Principal symbols. The phase symbols follow Volume 1, Part A, Section 9.2.

Symbol	Meaning	Units
$\theta_{CH}(f)$	One-way phase delay of the channel	rad
$\Delta\theta_{LO}(f)$	Relative phase difference of the two devices' carriers	rad
$\theta_{REFL}(f)$	Phase measured at the reflector's antenna	rad
$\theta_{INIT}(f)$	Phase measured at the initiator's antenna	rad
A_{REFL}, A_{INIT}	Measured carrier amplitudes at each device	—
PCT_{REFL}, PCT_{INIT}	Reported phase correction terms, complex	—
$H_2(f)$	Assembled two-way channel transfer function	—
φ_{app}	Phase the reflector applies to its outgoing carrier	rad
b	Applied-phase resolution of the reflector's transmitter	bits
T_{SW_IPT}	Antenna switch period declared for IPT operation	s
T_{IP2}	Interlude period, from the IPT time list when IPT is enabled	s
RPL	Reference power level accompanying the PCT values	dBm
σ_θ	Per-tone phase estimation standard deviation	rad
σ_Θ	Two-way phase standard deviation	rad

3 The mechanism

3.1 The ordinary construction

Volume 1, Part A, Section 9.2 sets out the baseline. With $\theta_{CH}(f)$ the one-way channel phase and $\Delta\theta_{LO}(f)$ the relative carrier phase between the devices, the phases measured at the two antennas are

$$\theta_{REFL}(f) = \theta_{CH}(f) + \Delta\theta_{LO}(f), \quad \theta_{INIT}(f) = \theta_{CH}(f) - \Delta\theta_{LO}(f), \quad (1)$$

and each device reports a phase correction term carrying that phase together with the measured amplitude,

$$\text{PCT}_{REFL}(f) = A_{REFL}(f) e^{i\theta_{REFL}(f)}, \quad \text{PCT}_{INIT}(f) = A_{INIT}(f) e^{i\theta_{INIT}(f)}. \quad (2)$$

The ranging entity forms the product, and the oscillator term cancels:

$$H_2(f) = \text{PCT}_{REFL}(f) \text{PCT}_{INIT}(f) = A_{CH2}(f) e^{i2\theta_{CH}(f)}. \quad (3)$$

Note where the cancellation happens: in a multiplication, performed wherever both terms have arrived.

3.2 The inline construction

With IPT enabled the reflector does not merely measure θ_{REFL} ; it applies that phase to its own outgoing carrier [Core 6.3, Vol. 6, Part A, Sec. 6.5]. The phase the initiator subsequently measures at its antenna therefore carries the reflector's measurement as well as the channel:

$$\theta_{INIT}(f) = \theta_{CH}(f) - \Delta\theta_{LO}(f) + \theta_{REFL}(f) = \theta_{CH} - \Delta\theta_{LO} + \theta_{CH} + \Delta\theta_{LO} = 2\theta_{CH}(f), \quad (4)$$

which is the relation given in Volume 1, Part A, Section 9.2. The unknown relative oscillator phase has cancelled — but it has cancelled *in the air*, between the reflector's transmitter and the initiator's receiver, rather than in an arithmetic product formed later.

Because the reflector has already handed over its phase, the term it reports carries none:

$$\text{PCT}_{REFL}(f) = A_{REFL}(f) e^{i0}, \quad (5)$$

and the assembled transfer function is unchanged from Eq. (3),

$$H_2(f) = A_{REFL}(f) e^{i0} \times A_{INIT}(f) e^{i\theta_{INIT}(f)} = A_{CH2}(f) e^{i2\theta_{CH}(f)}. \quad (6)$$

The identity that makes it work. Equation (4) is exact and requires no assumption beyond the channel reciprocity already assumed in Eq. (1). The reflector's measurement carries $+\Delta\theta_{LO}$ and the initiator's carries $-\Delta\theta_{LO}$; adding one to the other in the radio is algebraically the same operation as multiplying the two complex terms later. IPT is therefore not an approximation of ordinary phase-based ranging — it is the same computation, relocated.

3.3 What the initiator must still be given

Equation (6) still contains $A_{REFL}(f)$, and the reflector still reports it in the real part of its phase correction term. For a range extracted from the *slope* of $\arg H_2$ against frequency, the reflector's amplitude is irrelevant: it affects only the magnitude. For a range extracted from the *inverse transform* — the channel impulse response of the companion Mode-2 report —

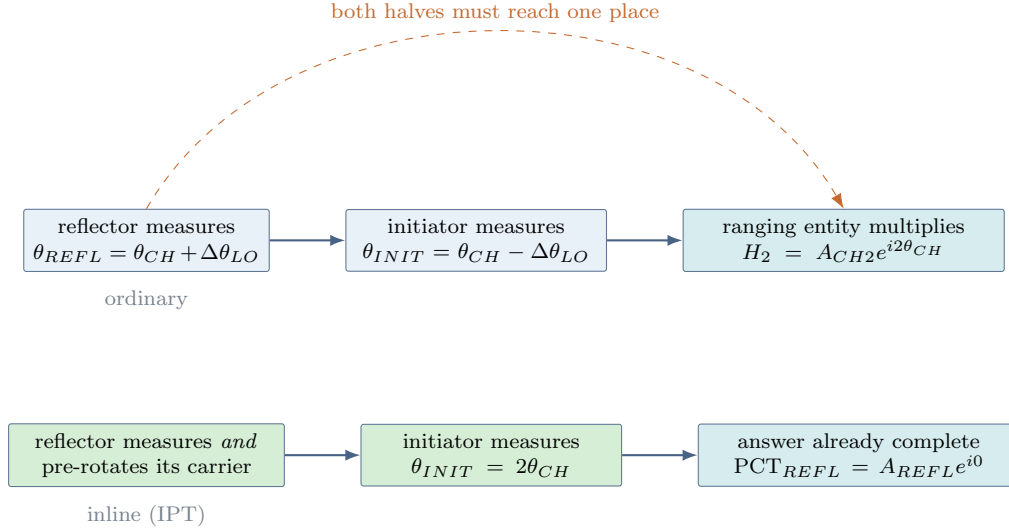


Figure 1: Where the two-way phase is formed. Ordinarily the cancellation of $\Delta\theta_{LO}$ happens in a product computed once both phase correction terms have arrived somewhere. With IPT it happens in the radio link itself, and the initiator's single measurement is already $2\theta_{CH}$.

the amplitude is a weighting, and discarding it changes the shape of the point-spread function and hence first-path detection.

A useful robustness property, and its limit. With IPT the initiator can compute a phase-slope range from its own measurements alone, so a lost or delayed reflector report degrades the result rather than preventing it. That is a real availability gain. It does not extend to transform-domain processing, which wants A_{REFL} as a weight — so an implementation that relies on first-path detection has not escaped the reverse data path, only made it non-blocking.

4 What IPT demands of the reflector

Equation (4) is one line of algebra. Implementing it is a transmitter requirement, and it is the substance of the feature.

4.1 A phase-commandable transmitter

The reflector must be able to set the phase of its transmitted carrier to a commanded value, per antenna path, per channel. Nothing in ordinary Channel Sounding requires this: a conventional reflector transmits an unmodulated tone whose absolute phase is arbitrary, because the ordinary construction never depends on it. Under IPT that arbitrary phase becomes the observable, and it must be placed deliberately.

Three implementation routes are common, and they differ in exactly the properties that Section 8 shows to matter.

Applied-phase resolution is a ranging specification. Section 8 quantifies it: at the worked configuration, eight bits of applied-phase resolution costs 2.5% of ranging precision and six bits costs 34%. A five- or six-bit RF phase shifter, entirely adequate for beam-steering, is not adequate here. This is the single most likely way for an IPT implementation to be quietly worse than the non-IPT implementation it replaced.

Table 3: Routes to a phase-commandable carrier, and the properties that bear on ranging.

Mechanism	Resolution	Settling	Principal risk
Direct digital synthesis of the tone at baseband, upconverted	Set by the numerically controlled oscillator word; easily 12 bits or more	Immediate, in the digital domain	Image and DC leakage rotate the effective phase
Programmable phase shifter in the RF path	Set by the shifter, often 5–8 bits	Device-dependent; must fit inside T_{SW_IPT}	Amplitude–phase coupling; state-dependent insertion loss
Phase-continuous synthesiser retune	Fine, but indirect	Slowest; may not fit a $1\,\mu s$ switch period	Settling transient inside T_{PM}

4.2 The real-time loop

The reflector must measure a path’s phase, compute the value to apply, program the transmitter, and have the carrier settled before its own tone slot for that path opens. The available margin follows from the Mode-2 step structure of the companion report: the reflector’s tone sequence begins after the initiator’s sequence, its ramp-down and the interlude, and the reflector transmits its antenna paths in the same order it measured them, so the margin is very nearly the same for every path:

$$T_{margin} \approx T_{RD} + T_{IP2}. \quad (7)$$

Table 4: Real-time margin available to the reflector’s measure–compute–apply loop, from Eq. (7) with $T_{RD} = 5\,\mu s$.

T_{IP2}	T_{margin}	Comment
$10\,\mu s$	$15\,\mu s$	The shortest configuration; demands a hardware-paced pipeline
$20\,\mu s$	$25\,\mu s$	Feasible for a tightly written firmware loop
$40\,\mu s$	$45\,\mu s$	Comfortable; the value used in the worked example
$80\,\mu s$	$85\,\mu s$	Generous
$145\,\mu s$	$150\,\mu s$	Mandatory value; ample, but lengthens the step and the reciprocal interval

The interlude is not free to lengthen. Relaxing T_{IP2} to buy computation time directly lengthens ΔT , the interval between the two reciprocal tone measurements, which the companion Mode-2 report shows enters the range as $c\epsilon\Delta T/2$ against the residual frequency offset. Moving from $T_{IP2} = 10\,\mu s$ to $145\,\mu s$ multiplies that term by roughly a factor of nine. An IPT implementation that needs a long interlude to close its loop is spending ranging accuracy to buy processing time, and should account for it explicitly rather than treating the interlude as a free parameter.

4.3 The phase accuracy requirement still applies

Section 6.5 states plainly that “a CS reflector shall satisfy the phase accuracy requirements” [Core 6.3, Vol. 6, Part A, Secs. 6.5 and 6.2]. Enabling IPT does not relax the conformance limits on the phase chain; it changes where in the chain the phase appears. The linear-regression

bounds of Section 6.2 — a fitted slope below $2\pi \times 1.7$ ns and residuals below 0.28 rad for 95% of values — must still be met, and under IPT the reflector’s applied-phase error is inside the loop those bounds constrain.

4.4 The tone extension slot is exempt

One explicit relaxation exists. When IPT is enabled, “there is no requirement to compensate the phase of the CS tone over the CS tone extension slot” [Core 6.3, Vol. 6, Part H, Sec. 4.3.3]. The extension slot exists as a noise and interference reference and as a duration-matching device, not as a ranging observable, so there is no measured phase to return for it. An implementation should not treat the extension slot’s phase as meaningful under IPT, and a receiver should not include it in a slope fit.

5 Timing parameters that exist only for IPT

Because the reflector’s transmitter now has work to do between slots, the specification gives IPT its own declarations for the two periods that bound that work [Core 6.3, Vol. 4, Part E; Vol. 6, Part H, Sec. 4.3].

Table 5: The IPT-specific timing declarations. Both are declared by the reflector alongside, and independently of, the ordinary values.

Parameter	Declared values	Purpose
T_{SW_IPT}	0, 1, 2, 4 or $10\mu s$ — the same set as T_{SW} , declared separately	The antenna switch period used when IPT is enabled, within which the new phase must also settle
T_{IP2} IPT time list	Bits for 10, 20, 30, 40, 50, 60 and $80\mu s$; $145\mu s$ remains mandatory and is not in the optional list	The interlude the reflector can support while closing its measure–compute–apply loop

Two observations follow from the value sets.

T_{SW_IPT} *buys no extra time by itself*. Its permitted values are identical to those of T_{SW} , so a reflector cannot ask for a longer switch period than the ordinary maximum of $10\mu s$. What the separate declaration provides is the ability to declare a *different* value — a reflector whose phase shifter settles in $4\mu s$ may support $T_{SW} = 1\mu s$ without IPT and require $T_{SW_IPT} = 4\mu s$ with it, and the two are negotiated independently. In the step-duration formulas, T_{SW} and T_{SW_IPT} are represented by the same symbol [Core 6.3, Vol. 6, Part H, Secs. 4.3.3 and 4.3.4], so the arithmetic of the companion Mode-2 and Mode-3 reports is unchanged; only which value is substituted differs.

The T_{IP2} IPT list *is where the real accommodation is*. A reflector that cannot close Eq. (7) at $10\mu s$ simply does not set that bit, and the configuration must then choose a longer interlude — with the accuracy consequence noted above.

6 Altered reporting

6.1 The zero-quadrature rule

Under IPT the reflector “shall only return PCT values where the imaginary (Q) part of each PCT value is zero”, and “the real (I) part of the PCT is used to convey the measured signal

amplitude” [Core 6.3, Vol. 6, Part A, Sec. 6.5]. Volume 6, Part H adds that the real part “shall be non-negative and should reflect the magnitude of measurement” [Core 6.3, Vol. 6, Part H, Sec. 4.6], and the host interface repeats the rule at the event level, noting that “on the initiator side, the Tone_PCT format remains unchanged” [Core 6.3, Vol. 4, Part E].

Table 6: What each device reports per tone, with and without IPT.

	Without IPT	With IPT
Reflector PCT	12-bit signed I and Q ; carries $A_{REFL}e^{i\theta_{REFL}}$	$Q \equiv 0$; I non-negative, carries A_{REFL} only
Initiator PCT	12-bit signed I and Q ; carries $A_{INIT}e^{i\theta_{INIT}}$	Unchanged; now carries $A_{INIT}e^{i2\theta_{CH}}$
Reference power level	8-bit signed dBm, one per subevent	Unchanged
Tone quality indicator	2-bit, per tone	Unchanged
Information in the reflector’s report	Phase and amplitude	Amplitude only

The field widths do not change: the reflector still emits a 24-bit `Tone_PCT` per tone, half of which is now known in advance to be zero. IPT therefore saves nothing in the host interface’s event size. What it saves is the *dependency*, not the bytes — the ranging-critical phase no longer has to arrive for a range to exist.

6.2 Amplitude and the reference power level

The reference power level is unaffected: it remains an 8-bit signed dBm value, one per subevent, against which the unitless I and Q are interpreted through

$$IQ_{dBm} = 20 \log_{10} \left(\frac{|IQ_{linear}|}{2048} \right) + RPL_{dBm} \quad (8)$$

[Core 6.3, Vol. 6, Part H, Sec. 4.6]. Under IPT the reflector’s $|IQ|$ is simply its I value, so Eq. (8) still yields a calibrated received power at the reflector — which is worth retaining, because it is the only remaining window into what the reflector actually saw.

7 Capability negotiation and configuration

IPT is optional, is a property of the reflector role, and is off unless deliberately switched on.

1. **Prerequisite feature.** IPT sits under the Channel Sounding Enhancement #1 LE feature; the IPT-specific time lists are meaningful only if that feature is supported [Core 6.3, Vol. 4, Part E].
2. **Capability declaration.** In its `Subfeatures.Supported` field the reflector sets bit 4, “IPT in the CS reflector”, and it populates the two IPT time lists tabulated in Appendix A. A device that does not support IPT in the reflector role sets T_{SW_IPT} to zero [Core 6.3, Vol. 4, Part E].
3. **Link-layer exchange.** The declaration travels between the devices in the version-2 `LL_CS_CAPABILITIES` request and response PDUs [Core 6.3, Vol. 6, Part B].
4. **Configuration.** IPT is enabled by bit 0, “IPT is enabled in the CS reflector”, of the `CS_Enhancements` configuration octet [Core 6.3, Vol. 4, Part E].

5. **Default.** The specification is explicit that the parameter “shall be set by default to not enabled and may be set to enabled if support for IPT was indicated in the peer’s LL_CS_CAPABILITIES_REQ [v2] or LL_CS_CAPABILITIES_RSP [v2] PDU” [Core 6.3, Vol. 6, Part B].

Both roles must know. The reflector changes what it transmits; the initiator must know that the reflector’s reported quadrature is meaningless and that its own measurement is already the two-way phase. An initiator that applies the ordinary product of Eq. (3) to an IPT reflector’s report multiplies by a purely real number and obtains $A_{CH2}e^{i2\theta_{CH}}$ — which happens to be correct. An initiator that instead *adds* unwrapped angles, as the companion Mode-2 report warns against, adds zero and also obtains the right answer. The dangerous case is the reverse: an initiator that assumes IPT when it is not enabled will treat $\theta_{CH} - \Delta\theta_{LO}$ as $2\theta_{CH}$ and produce a confidently wrong range contaminated by the oscillator phase. The configuration state must be read, not inferred from a zero quadrature.

8 Error analysis

8.1 IPT is thermally free

It is worth establishing this carefully, because the intuition that returning a measurement through the radio must cost something is wrong.

Without IPT, the two-way phase is the sum of two independent estimates. Writing ε_R and ε_I for the estimation errors at the reflector and initiator, each of variance σ_θ^2 ,

$$\arg H_2 = 2\theta_{CH} + \varepsilon_R + \varepsilon_I, \quad \sigma_\Theta^2 = 2\sigma_\theta^2. \quad (9)$$

With IPT, the reflector applies its *estimate* $\theta_{REFL} + \varepsilon_R$ to the carrier. The initiator then measures that, with its own error:

$$\theta_{INIT} = \theta_{CH} - \Delta\theta_{LO} + (\theta_{REFL} + \varepsilon_R) + \varepsilon_I = 2\theta_{CH} + \varepsilon_R + \varepsilon_I, \quad (10)$$

which is term for term the same as Eq. (9).

The same two errors, combined the same way. IPT does not average, duplicate or amplify the phase estimates; it relocates the addition. Both constructions accumulate exactly one reflector-side and one initiator-side phase error, so the thermal precision of the companion Mode-2 report — $\sigma_\Theta = \sqrt{2}\sigma_\theta$, giving 4.065 mm over the full 72-channel raster at 20 dB — carries across to IPT unchanged. Any claim that IPT is inherently more or less precise is mistaken; what differs is the hardware term below.

8.2 The term IPT adds

What IPT does introduce is an error that does not exist otherwise: the reflector’s applied phase is not exactly the phase it intended. Write η for that applied-phase error, so that Eq. (10) becomes $2\theta_{CH} + \varepsilon_R + \varepsilon_I + \eta$.

The dominant contribution to η in most implementations is the resolution of whatever sets the carrier phase. For a uniform quantiser of b bits over 2π ,

$$\sigma_\eta = \frac{2\pi}{2^b\sqrt{12}}. \quad (11)$$

Table 7: Cost of applied-phase resolution, evaluated on the full 72-channel CS raster at $\gamma = 20$ dB and $T_{PM} = 10 \mu s$, where the baseline two-way phase noise is 1.812° and the baseline ranging precision is 4.065 mm.

b	Applied-phase LSB	σ_η	Its own σ_d	Combined σ_d
6	5.625°	1.624°	3.643 mm	5.458 mm (+34.3%)
7	2.812°	0.812°	1.821 mm	4.454 mm (+9.6%)
8	1.406°	0.406°	0.911 mm	4.166 mm (+2.5%)
9	0.703°	0.203°	0.455 mm	4.090 mm (+0.6%)
10	0.352°	0.101°	0.228 mm	4.071 mm (+0.2%)
12	0.088°	0.025°	0.057 mm	4.065 mm (+0.0%)

Eight bits is the knee. Below eight bits the applied-phase quantiser becomes comparable to the thermal floor and then dominates it; at and above eight bits it is a small correction. A direct-digital-synthesis implementation is comfortably past the knee without trying. An implementation built on a coarse RF phase shifter is not, and should either interpolate between shifter states or accept a quantified loss — but not leave it uncharacterised.

8.3 Errors that behave differently under IPT

Table 8: How each error mechanism behaves with and without IPT.

Mechanism	Without IPT	With IPT	Verdict
Thermal phase noise	$\varepsilon_R + \varepsilon_I$	$\varepsilon_R + \varepsilon_I$	unchanged
Applied-phase quantisation	absent	σ_η , Eq. (11)	IPT worse
Applied-phase settling inside T_{PM}	absent	slot-dependent bias	IPT worse
Reflector PCT quantisation (12-bit)	1.4×10^{-4} rad	not in the phase path	IPT better
Reflector amplitude-phase coupling	absent	shifter state changes insertion phase	IPT worse
Residual frequency offset over ΔT	$c\epsilon\Delta T/2$	unchanged, but T_{IP2} may be longer	often IPT worse
Loop phase response $\Phi_{loop}(f)$	calibrated at the ranging entity	reflector's half must be right at transmit time	unchanged in principle
Reverse-path loss or delay	range unavailable	phase-slope range still available	IPT better

Two rows deserve expansion. The reflector's 12-bit phase correction term quantisation, which the companion Mode-2 report puts at 0.003 cm of range, disappears from the phase path entirely under IPT — a small genuine gain, and the only one in the table. And amplitude-phase coupling in a switched phase shifter is a mechanism with no analogue in ordinary operation: changing the applied phase changes the insertion loss, which changes the amplitude the initiator measures, which perturbs a transform-domain weighting. It is calibratable, per state, and should be.

9 What IPT permanently removes

The costs in Section 8 are quantities. This section is about something that cannot be put in a budget table, and it is the strongest argument against enabling IPT casually.

Without IPT, the ranging entity holds θ_{REFL} and θ_{INIT} as separate numbers. With IPT, θ_{REFL} has been folded into the radio signal and the field that would have carried it is required to be zero. It is not compressed, not approximated — it is gone, and nothing downstream can recover it.

Table 9: Capabilities that depend on holding the two phases separately.

Capability	Why it needs both phases	Status under IPT
Attributing error to a device	The two contributions are only separable when reported separately	Lost; a bad range cannot be blamed
Re-processing a stored procedure	Better algorithms need the raw observables	Lost; only the fused result was ever recorded
Per-device calibration from field data	Requires isolating one device’s phase response	Lost
Detecting a faulty reflector	An implausible θ_{REFL} is visible	Lost; a wrong applied phase looks like channel
Asymmetric antenna diagnostics	Per-path phase at each end	Halved
Independent cross-check	Two numbers can disagree; one cannot	Lost unless Mode-3 supplies it

An IPT range carries no evidence of how it was formed. Every error in the reflector’s phase estimate and in its applied phase arrives at the initiator disguised as channel phase, and therefore as distance. A reflector whose phase shifter has drifted, whose calibration table is stale, or whose measurement window has slipped will produce ranges that are wrong by a fixed amount and entirely self-consistent — smooth across frequency, low in residual, and indistinguishable from a target at a different distance. Ordinary operation would have exposed the same fault as a disagreement between two reported numbers. This is the reason to keep a non-IPT mode available for commissioning and field diagnosis even in a product that runs IPT in service.

10 Integrity, and why IPT belongs with Mode-3

The companion Mode-2 report establishes that a tone is fully predictable and therefore that phase-based ranging is not, on its own, a distance-bounding measurement. IPT does not change that conclusion, but it does move the trust boundary in a way worth stating precisely.

What does not change. A malicious reflector could always lie. Without IPT it lies in its reported phase correction terms; with IPT it lies in the phase it applies to its carrier. Neither is detectable from the phase observable alone, because the initiator’s own measurement is contaminated by $\Delta\theta_{LO}$ in the first case and is by construction the fused result in the second. IPT is therefore not a security regression against a hostile peer.

What does change. Against a *faulty* rather than hostile peer, ordinary operation gives the ranging entity two numbers that can be checked against each other and against physics. IPT gives it one. The practical consequence is not a weaker security posture but a weaker fault-detection posture, and the two are easy to confuse.

Mode-3 restores what IPT removes. A Mode-3 step produces a round-trip time and a phase observable from the same step, on the same channel, through the same hardware — the subject of the companion Mode-3 report. That round-trip time is measured independently of any phase the reflector applies, so it remains a valid check on an IPT phase result. The per-step consistency residual $\delta_k = \hat{d}_{\text{RTT},k} - \hat{d}_{\Theta,k}$, and the 3σ gate of 38.7 cm derived there, are exactly the auditing mechanism that Section 9 says IPT otherwise destroys. IPT and Mode-3 are therefore complementary rather than merely compatible, and an IPT deployment that also needs assurance should prefer Mode-3 over Mode-2.

11 When to enable IPT

Table 10: Summary of the trade.

Dimension	Effect of enabling IPT	Magnitude
Thermal ranging precision	None	σ_{Θ} identical
Applied-phase quantisation	New error	+2.5% at 8 bits, +34% at 6
Reflector PCT quantisation	Removed from the phase path	−0.003 cm
Dependence on the reverse data path	Removed for a slope-fit range	Qualitative, large
Latency to a usable range	Removed the reverse-path wait	Implementation-dependent
Reflector transmitter complexity	Phase-commandable carrier required	Substantial
Reflector real-time budget	Must close in $T_{RD} + T_{IP2}$	As little as 15 μs
Host-interface event size	Unchanged	Field widths identical
Fault attribution and re-processing	Lost	Absolute

A decision rule. Enable IPT when the initiator must produce a range without waiting on the reflector’s report — a low-latency access-control gesture, a tracker that may lose the reverse link, an initiator that is the only device with the ranging algorithm — *and* the reflector’s transmitter can command phase to at least eight bits and settle inside the negotiated T_{SW_IPT} *and* the deployment can tolerate the loss of fault attribution, or restores it with Mode-3. Leave IPT disabled when the reflector’s phase is wanted for calibration, diagnosis or re-processing; when the reflector’s phase-setting hardware is coarse; or when the only way to close the real-time loop is a long interlude, since that spends more accuracy through $c\epsilon\Delta T/2$ than IPT was worth.

12 Worked example

Take the configuration of the companion Mode-2 report — antenna configuration index 0 so that $N_{AP} = 1$, $T_{SW} = 2\mu s$, $T_{PM} = 10\mu s$, $T_{RD} = 5\mu s$, the full 72-channel CS raster spanning 74 MHz, a per-tone signal-to-noise ratio of 20 dB, a true separation of 10.000 m — with $T_{IP2} = 40\mu s$ and IPT enabled on the reflector. The tone sequence is then $(T_{SW} + T_{PM})(N_{AP} + 1) = 24\mu s$ and the reciprocal measurement interval is $\Delta T = 69\mu s$.

12.1 The observable

The reflector measures $\theta_{REFL}(f_k)$ on each channel and applies it to its own carrier. The initiator measures, per Eq. (4), $\theta_{INIT}(f_k) = 2\theta_{CH}(f_k)$ directly. At 2 440 MHz and 10.000 m this is 4.896351 rad modulo 2π , advancing by 24.0166° per megahertz — numerically identical to the two-way phase the Mode-2 report assembles from two terms, as it must be.

The reflector reports $PCT_{REFL} = A_{REFL}e^{i0}$: quadrature zero, real part non-negative and proportional to the received amplitude, interpreted against the subevent reference power level through Eq. (8).

12.2 The real-time budget

From Eq. (7), the reflector has $T_{RD} + T_{IP2} = 5 + 40 = 45 \mu s$ between the end of a path's measurement and the start of its own transmission on that path. Within that window it must complete the phase estimate over the captured T_{PM} , compute the value to apply, and program the transmitter; the carrier must then be settled within the negotiated T_{SW_IPT} before its tone slot opens. At the shortest permitted interlude the same budget would be $15 \mu s$.

12.3 Precision

Table 11: Ranging precision at the worked configuration, with the applied-phase term added to the companion Mode-2 report's budget. All other rows are unchanged by IPT.

Contribution	Phase	Range	Character
Loop group-delay residual (0.1 ns)	—	1.4990 cm	bias
Thermal noise, 72 channels	1.81° per tone	0.4065 cm	random
Oscillator phase noise (0.5° rms)	0.71° per tone	0.1586 cm	partly correlated
Applied-phase quantisation, 8 bits	0.41° per tone	0.0911 cm	random, IPT-only
Mode-0 residual, 0.05 ppm at $\Delta T = 69 \mu s$	—	0.0517 cm	bias
FAE table quantisation at ΔT	—	0.0162 cm	bias, per channel
Initiator PCT quantisation, 12-bit	0.008°	0.0026 cm	random
Reflector PCT quantisation	—	0	removed by IPT
Root-sum-square (excluding multipath)	—	1.5648 cm	

The comparison is the point of the table. Repeating the same budget *without* IPT — the applied-phase row removed, the reflector's 12-bit phase correction term restored to the phase path — gives 1.5621 cm. Enabling IPT at eight bits therefore costs 0.0027 cm, or $27 \mu m$ of root-sum-square uncertainty: not measurable against a budget whose leading term is a calibration residual fifty times larger. At six bits the applied-phase row becomes 0.3643 cm and the total rises to 1.6040 cm, which is a real if still modest degradation. IPT is, in precision terms, free — provided the phase is applied well.

Table 12: Deterministic values for the worked example.

Quantity	Value	Unit
$N_{AP} / T_{SW} / T_{PM} / T_{RD} / T_{IP2}$	1 / 2 / 10 / 5 / 40	—, μs
Tone sequence $(T_{SW} + T_{PM})(N_{AP} + 1)$	24	μs
Reciprocal interval ΔT	69	μs
Channels / pitch / span	72 / 1 / 74	—, MHz, MHz
Reflector real-time margin $T_{RD} + T_{IP2}$	45	μs
Same at the shortest interlude	15	μs
Two-way phase at f_c , modulo 2π	4.896351	rad
Two-way phase step per 1 MHz	24.0166	deg
Baseline σ_Θ	1.812	deg
Baseline σ_d , 72 channels	4.065	mm
Applied-phase σ_η , 8 bits	0.406	deg
Combined σ_d , 8 bits	4.166	mm
Combined σ_d , 6 bits	5.458	mm
Budget RSS with IPT, 8 bits	1.5648	cm
Budget RSS with IPT, 6 bits	1.6040	cm
Budget RSS without IPT, same configuration	1.5621	cm
Cost of IPT at 8 bits	0.0027	cm

13 Verification and conformance-oriented test plan

The Mode-2 verification plan of the companion report applies unchanged to the phase chain. The following are specific to IPT.

1. **Zero-quadrature conformance.** With IPT enabled, verify that every reflector `Tone_PCT` has $Q = 0$ exactly and $I \geq 0$, on every channel, every antenna path and the tone extension slot [Core 6.3, Vol. 6, Part A, Sec. 6.5; Vol. 6, Part H, Sec. 4.6]. A non-zero quadrature is a conformance failure, not a rounding artefact.
2. **The inline identity.** Drive the reflector with a reference transmitter at a known channel phase and confirm that the initiator's measured θ_{INIT} equals $2\theta_{CH}$ to within the documented residual — that is, that Eq. (4) holds and the local-oscillator term has genuinely cancelled. Repeat with a deliberately offset reference oscillator and confirm the result is invariant: this is the test that proves the cancellation is real and not a coincidence of one setup.
3. **Applied-phase linearity and resolution.** Command the reflector's applied phase across a full 2π in fine steps and measure the achieved phase. Extract the effective number of bits, the integral and differential non-linearity, and any amplitude-phase coupling. Compare the measured σ_η against Table 7 and confirm the implementation sits at or above the eight-bit knee.
4. **Settling inside T_{SW_IPT} .** Sweep the applied-phase step size — including the worst case of a near- π transition between consecutive antenna paths — and verify the carrier is settled before T_{PM} opens at the negotiated T_{SW_IPT} . A residual settling transient appears as a slot-dependent phase bias, which the companion Mode-2 report shows is indistinguishable from an antenna calibration error.
5. **Real-time margin.** Verify the measure–compute–apply loop closes at every T_{IP2} value the reflector declares, including the shortest. Instrument the loop and confirm the margin

- against Eq. (7); a reflector that declares $10\mu s$ and does not in fact close in $15\mu s$ will emit stale phases with no error indication.
6. **Stale-phase behaviour.** Deliberately delay the applied phase by one antenna path and confirm the resulting range error is detected by the Mode-3 consistency gate. This documents the failure mode that Section 9 says is otherwise invisible.
 7. **Capability and configuration.** Verify that IPT is off by default; that it can be enabled only when the peer declared support; that T_{SW_IPT} is reported as zero when the reflector role does not support IPT; and that the negotiated T_{IP2} is drawn from the IPT list rather than the ordinary one [Core 6.3, Vol. 4, Part E; Vol. 6, Part B].
 8. **Mismatched assumption.** Configure the reflector without IPT while the initiator assumes it, and confirm the resulting range is grossly wrong and is caught — by the Mode-3 gate where available, or by a plausibility check where not. This is the dangerous asymmetry identified in Section 7.
 9. **Tone extension slot.** Confirm the reflector does not compensate phase over the extension slot and that the initiator excludes it from any slope fit or transform.
 10. **Equivalence regression.** Run the same geometry with IPT enabled and disabled and confirm the ranges agree to within the difference predicted by Table 11. A systematic offset between the two modes indicates an applied-phase sign or reference-plane error at the reflector.
 11. **Phase accuracy conformance.** Confirm the Section 6.2 limits — fitted slope below $2\pi \times 1.7\text{ ns}$, residuals below 0.28 rad for 95% of values — are met with IPT enabled, since Section 6.5 does not relax them [Core 6.3, Vol. 6, Part A, Secs. 6.2 and 6.5].
 12. **Reverse-path independence.** Suppress the reflector's report entirely and confirm the initiator still produces a phase-slope range, and that it correctly degrades or disables transform-domain first-path processing that needs A_{REFL} .

13.1 Minimum acceptance records

Retain everything the Mode-2 plan requires, and in addition: the negotiated IPT state, T_{SW_IPT} and T_{IP2} ; the measured applied-phase transfer characteristic with its effective number of bits and amplitude-phase coupling; the settling characterisation against T_{SW_IPT} ; the measured real-time margin at each declared T_{IP2} ; the IPT-enabled and IPT-disabled range pair from the equivalence regression; and, where Mode-3 is used, the consistency residual distribution with IPT enabled.

14 Normative behaviour versus implementation choice

Four conclusions that prevent common design errors. (1) IPT is thermally free — Eq. (10) accumulates exactly the same two phase errors as ordinary operation — so any precision difference observed in practice is a hardware defect, not a property of the feature. (2) Applied-phase resolution is a ranging parameter: eight bits costs 2.5%, six bits costs 34%. (3) Buying real-time margin with a longer interlude spends ranging accuracy through $c\epsilon\Delta T/2$, and can easily cost more than IPT saves. (4) The reflector's phase is not compressed but destroyed; fault attribution, re-processing and per-device calibration all go with it, and only a Mode-3 round-trip time restores an independent check.

Table 13: Boundary between specified behaviour and implementation design for IPT.

Bluetooth-defined behaviour	Implementation-specific design
That the reflector adjusts its CS tone phase so the reported PCT has zero phase [Core 6.3, Vol. 6, Part A, Sec. 6.5]	How the carrier phase is commanded: direct digital synthesis, RF phase shifter, synthesiser retune
$Q = 0$; I non-negative and conveying amplitude [Core 6.3, Vol. 6, Part H, Sec. 4.6]	Amplitude scaling and its relation to the reference power level
That phase accuracy requirements still apply [Core 6.3, Vol. 6, Part A, Sec. 6.2]	Applied-phase resolution, linearity and calibration
T_{SW_IPT} and the T_{IP2} IPT time list, and their declaration	Which values are supported, and the real-time architecture that justifies them
Exemption of the tone extension slot from phase compensation	Whether the extension slot is measured at all
Capability bit, enable bit, and the default-off rule [Core 6.3, Vol. 4, Part E; Vol. 6, Part B]	Policy for when to request IPT
The phase relation of Eq. (4) [Core 6.3, Vol. 1, Part A, Sec. 9.2]	The entire ranging algorithm downstream of it

15 Conclusions

Inline PCT Transfer is a relocation, not an improvement. The two-way phase that ordinary phase-based ranging assembles by multiplying two reported complex terms is instead assembled in the radio link, by having the reflector pre-rotate its carrier by the phase it has just measured. The algebra is exact, the specification states it in one line, and the resulting observable is identical: the initiator’s own measurement becomes $2\theta_{CH}$, and the reflector’s report is left carrying amplitude alone with its quadrature required to be zero.

Because the same two phase estimates are combined in the same way, IPT is thermally free. Its genuine costs are elsewhere. It requires a reflector transmitter whose carrier phase can be commanded per antenna path and settled inside a switch period, and this report quantifies what that costs when done coarsely: eight bits of applied-phase resolution is the knee, below which the quantiser overtakes the thermal floor. It requires a measure–compute–apply loop closed in as little as fifteen microseconds, and the tempting remedy — a longer interlude — is itself paid for in ranging accuracy through the residual-frequency term. And it permanently removes the reflector’s phase as a recorded quantity, which ends fault attribution, re-processing and per-device calibration from field data.

The feature therefore earns its place in a specific architecture: one where the initiator must have a range without waiting on the reverse data path, where the reflector’s radio is good enough to place phase precisely, and where either the loss of diagnostic redundancy is acceptable or Mode-3 is used to restore it. Those conditions are neither rare nor universal, which is presumably why IPT is optional, negotiated, and off by default.

A Capability and configuration fields

Two notes. The IPT enable is a *configuration* parameter, not a per-procedure hint: it holds for the CS procedure and both devices must agree on it, which is why Section 7 insists it be read rather than inferred. And the prerequisite Channel Sounding Enhancement #1 feature governs whether the IPT time lists carry meaning at all, so a capability parser must check

Table 14: Host-interface and link-layer fields governing IPT. Field names follow Core 6.3, Volume 4, Part E and Volume 6, Part B.

Field	Where	Meaning
Subfeatures_Supported bit 4	Capability	“IPT in the CS reflector”
T_IP2_IPT_Times_Supported	Capability, 2 octets	Optional interlude values, bits for 10–80 μs ; 145 μs mandatory and not listed
T_SW_IPT_Times_Supported	Capability, 1 octet	0x00, 0x01, 0x02, 0x04 or 0x0A microseconds; zero if the reflector role does not support IPT
CS_Enhancements bit 0	Configuration	“IPT is enabled in the CS reflector”
LL_CS_CAPABILITIES [v2]	Link layer	Request/response PDUs carrying the declaration
Tone_PCT , reflector	Result	Q shall be 0; I conveys amplitude
Tone_PCT , initiator	Result	Format unchanged

the feature before interpreting them.

B Numerical test vectors

- E1. Inline identity.** With $\theta_{CH} = 0.7\text{rad}$ and $\Delta\theta_{LO} = 1.3\text{rad}$, confirm that ordinary operation gives $\theta_{REFL} = 2.0$, $\theta_{INIT} = -0.6$ and $\arg(\text{PCT}_{REFL}\text{PCT}_{INIT}) = 1.4 = 2\theta_{CH}$; and that with IPT the initiator alone measures $\theta_{INIT} = 1.4$. Repeat for $\Delta\theta_{LO} \in \{0, \pi/2, \pi, 3\pi/2\}$ and confirm invariance.
- E2. Zero quadrature.** Confirm every reflector Tone_PCT has $Q = 0$ and $I \geq 0$ across all 72 channels and both tone slots at $N_{AP} = 1$.
- E3. Two-way phase.** At $d = 10.000\text{m}$ and 2440MHz , confirm the initiator's measured phase is 4.896351rad modulo 2π and advances by 24.0166° per megahertz — the same values the companion Mode-2 report obtains from two terms.
- E4. Applied-phase resolution sweep.** Quantise the applied phase to $b = 6, 7, 8, 9, 10, 12$ bits and confirm the resulting ranging standard deviations of $5.458, 4.454, 4.166, 4.090, 4.071$ and 4.065mm .
- E5. Budget equivalence.** At $b = 8$ confirm the assembled budget of 1.5648cm against the non-IPT value of 1.5621cm for the same configuration, a difference of 0.0027cm .
- E6. Real-time margin.** For $T_{IP2} \in \{10, 20, 40, 80, 145\}\mu\text{s}$ confirm the available margins of $15, 25, 45, 85$ and $150\mu\text{s}$ from Eq. (7).
- E7. Interlude cost.** Confirm that moving T_{IP2} from 10 to $145\mu\text{s}$ at $T_{SW} = 2\mu\text{s}$, $T_{PM} = 10\mu\text{s}$, $N_{AP} = 1$ raises ΔT from 39 to $174\mu\text{s}$ and the residual-frequency term at 0.05ppm from 0.029 to 0.130cm .
- E8. Mismatch.** Simulate an initiator assuming IPT against a reflector without it, and confirm the range error equals the contribution of $\Delta\theta_{LO}$ and is flagged.

References

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- [2] Bluetooth SIG, *Bluetooth Core Specification, Version 6.3*, Vol. 6, Part A, “Radio Physical Layer Specification,” Sec. 6.5 “Inline PCT Transfer,” Sec. 6.2 “Phase measurement accuracy,” Sec. 6.3 “Frequency actuation error correction” and Sec. 5.3 “Antenna Switching for Channel Sounding,” adopted 2026.
- [3] Bluetooth SIG, *Bluetooth Core Specification, Version 6.3*, Vol. 6, Part H, “Channel Sounding,” Secs. 4.3.3, 4.3.4 and 4.6, adopted 2026.
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